Data Product Specification of AIP Data Set – The People's Republic of China

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Published	2026-01-15
Language	English
Extent of the data product	Within the FIRs of the People's Republic of China
Topic category	Transportation
Keywords	AIP

Overview of the data product
The aeronautical data of the AIP data set of China are collected and published according to ICAO Annex 15, 16th Edition requirements. The AIP data set contains the following aeronautical data features: ● Aerodrome data ● Airspace data ● ATS and other routes data ● Navaid data ● Other information data

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Revision Record		
Document Version	**Revision Date**	**Revision Content**
1.0.0-beta0	2026-01-15	Initial release for trial operation.

1.About the data product specification	
Title	Data Product Specification of AIP Data Set - The People's Republic of China
This Version	https://www.eaipchina.cn
Latest Version	https://www.eaipchina.cn
Published	2026-01-15
Updated	NIL
Language	English
Contact	**Organisation:** Aeronautical Information Service Center of Air Traffic Management Bureau, Civil Aviation Administration of China **Address:** No.9 Xiedao West Road, Chaoyang District Beijing 100018, The People's Republic of China **Phone / Fax:** 86-10-57803699/86-10-57803650 **Email:** aipchina@atmb.net.cn **Web site:** https://www.aischina.com
Web location	https://www.eaipchina.cn
Format	PDF
Maintenance	The data product specification is updated regularly and reviewed at least once every year.
Handling restrictions	For aviation-related purposes only, including aviation operations, air traffic management, flight planning, etc. Unauthorized use in other fields is prohibited.
Terms and definitions	See ICAO Annex 15, 16th Edition and PANS-AIM (Doc 10066), 1st Edition.
Abbreviations	See ICAO Annex 15, 16th Edition and PANS-AIM (Doc 10066), 1st Edition.
2.Identification and purpose of the data product	
Official title	AIP data set of The People's Republic of China
Alternative title	AIP data set of China
Abstract	The aeronautical data of the AIP data set of China are collected and published according to ICAO Annex 15, 16th Edition requirements. The AIP data set contains the following aeronautical data features: ● Aerodrome data ● Airspace data ● ATS and other routes data ● Navaid data ● Other information data
Purpose	The purpose of the data product is given in ICAO PANS-AIM, Chapter 5.3.3.1 which provides the use of the data. It is the responsibility of the users to determine if the data product meets their needs.
Topic category	Transportation

Keywords	AIP, Airport, Airspace, Routes, Navaid
Spatial representation	Vector
Spatial resolution	Not applicable
Supplemental information	NIL
Restrictions	**Use limitations**: For aviation use only **Access restrictions**: Open to Certified Aeronautical Information Users **Usage restrictions**: Need to Comply with Relevant Regulations of Civil Aviation Administration of China **Security restrictions**: Unclassified
Extent	Within the FIRs of the People's Republic of China
3.Scopes	
General scope	
Scope id	General scope
Level	Dataset
Level name	General scope
Level description	The general scope level defines the specifications which are applicable to AIP Data Set of the People's Republic of China.
Extent	Within the FIRs of the People's Republic of China
Coverage	Not applicable
4.Data content and structure	
General scope	
Narrative description	The data model of the AIP Data Set of the People's Republic of China follows the Aeronautical Information Exchange Model (AIXM) 5.1.1 version specification, mapping the AIP data set subjects and properties specified in PANS-AIM to AIXM features, attributes, and associations, ensuring the data structure complies with international general standards and supports the interaction and integration of aeronautical information data with other countries and regions.
Application schema	See Annex A
Feature catalogue	See Annex B
5.Reference system	
General scope	
Spatial reference system	**Horizontal system**: World Geodetic System of 1984 geodetic reference datum (WGS-84) **Vertical system**: National vertical datum 1985
Temporal reference system	Gregorian Calendar, UTC
6.Data quality requirements	
General scope	
Requirement 1	**Data quality element**: Accuracy **Data quality measure**: The current accuracy is equivalent to that of China's AIP.
Requirement 2	**Data quality element**: Resolution

	Data quality measure: The current resolution is equivalent to that of China's AIP.
Requirement 3	**Data quality element**: Assurance (Integrity) **Data quality measure**: The current integrity requirements is equivalent to that of China's AIP. The procedures for processing aeronautical data have been setup to meet the integrity requirements.
Requirement 4	**Data quality element**: Traceability **Data quality measure**: All actions applied to the aeronautical data have been recorded.
Requirement 5	**Data quality element**: Timeliness **Data quality measure**: Timeliness is assured by providing the start and end time of all aeronautical data according to the temporality concept of AIXM.
Requirement 6	**Data quality element**: Completeness **Data quality measure**: All features and attributes are expressed according to the AIXM model.
7.Data capture and production	
General scope	
Description	Data has been captured according capture rules for aeronautical data are based on ICAO Doc for the Origination of Aeronautical Data.
Guide	NIL
Inclusion criteria	The data set includes all subjects and properties that are listed in PANS-AIM Chapter 5.3.3.1 "AIP Data set" plus additional data elements marked with #AIP-DS# in PANS-AIM, Appendix 2.
Data acquisition and processing	Not applicable
8.Data maintenance	
General scope	
Description	The AIP Dataset will be updated and published in accordance with the requirements specified in ICAO Annex 15. Baseline data set will be updated every AIRAC cycle (consistent with AIP). Supplementary and corrigenda will be published as needed.
Frequency	Periodic (every 28 days, consistent with AIP) and as Needed (for supplementary, corrigenda, etc.)
User defined	Not applicable
9.Portrayal rules	
General scope	
Portrayal rules	Not applicable
10.Data delivery methods	
General scope	
Format name	Aeronautical Information Exchange Model (AIXM)
Format version	AIXM 5.1.1
Format specification	AIXM 5.1.1 Specification

	(https://aixm.aero/page/aixm-511-specification)
File structure	http://www.aixm.aero/schema/5.1.1/AIXM_Features.xsd
Language	eng
Character set	UTF-8
Units of delivery	Dataset (See Annex C)
Transfer size	Not applicable
Medium name	Not applicable
Other delivery	Not applicable
Service property	Not applicable
11.Metadata	
General scope	
Specification	**Title:** ISO 19115:2003, Geographic information – Metadata **Date:** 2003
Encoding	**Title:** ISO 19139:2007, Geographic information – Metadata – XML schema implementation **Date:** 2007
Metadata elements	The following metadata is provided: ● Name of provider ● Date and time when provide ● Validity of the data set ● Indication that this is an AIRAC update ● Limitations on use
12.Additional information	
General scope	
Additional information	■ At the point of release the file contains information available in the public domain. The content carries the same data quality statement and limitations as the content of the official China AIP. ■ Checklist will be released and updated synchronously with AIP dataset issues. Refer to Annex D for relevant structure.

Annex A - Application schema

The application schema complies with the specifications of the Aeronautical Information Exchange Model (AIXM) Version 5.1.1, for details, please refer to the official documentation (https://aixm.aero/page/aixm-511-specification).

The extended or differentiated contents are as follows:

1. Due to system implementation constraints, this dataset **does not guarantee** that the sequenceNumber and correctionNumber contained in the TimeSlice are incremented consecutively upon release (i.e., the value difference of sequenceNumber or correctionNumber between two adjacent released TimeSlices is not necessarily 1, and **the interval is not fixed**). However, it shall follow the core rule that the sequenceNumber or correctionNumber corresponding to the newly released version is **greater than** the corresponding values of all previous versions. Please note this when using the dataset.

Annex B - Feature catalogue

The data catalogue with a description of all features, their properties and sub-properties, data quality requirement and the mapping to AIXM 5.1.1 is provided in the following attached files:

- Dataset Publication Scope and Mapping to AIXM5.1.1 of AIP Data Set-China _v1.0.0-beta0.xlsx

Annex C - Naming Conventions for AIP Data Set Packages and Files

1. Overview

China's AIP datasets are organized into data packages based on release content, ensuring standardized management, transmission, and identification of aeronautical data. These packages are categorized into **Baseline Data Packages** and **Supplementary Data Packages** to distinguish between regular, comprehensive data releases and ad-hoc supplementary updates. This annex specifies the naming rules for both data packages and their internal data files, ensuring consistency and clarity across the data lifecycle.

2. Data Package Organization

2.1 Baseline Data Package

A Baseline Data Package contains the complete set of valid aeronautical data for a specific period, including all current Baseline data, corresponding differential data (to track changes from previous versions), and checksum files for data integrity verification.

- **Naming Convention:**

  CN_AIP-DS_[Effective Time]_[Publication Number]_[Version Number].zip

- **Component Specifications:**
  - **[Effective Time]**: The time when the data in the package becomes valid, formatted as **EFF[UTC]** (e.g., EFF202501221600 for 16:00 UTC on January 22, 2025).
  - **[Publication Number]**: For AIRAC compliant releases, this follows the format **AIRAC[Amendment Number]** (e.g., AIRAC2501 for the first AIRAC cycle of 2025). For non-AIRAC releases, it is defined by business requirements.
  - **[Version Number]**: Indicates the release version of the package, formatted as **V[Version Number]**. The initial release of a package is uniformly "V0".

- **Example:**

  CN_AIP-DS_EFF202501221600_AIRAC2501_V0.zip

2.2 Supplementary Data Package

A Supplementary Data Package contains temporary updates of the data set with a validity of three months or longer

(the equivalent of AIP SUP data), along with corresponding checksum files.

- **Naming Convention:**

  CN_AIP-DS_[SUP Number]_[Version Number].zip

- **Component Specifications:**
  - **[SUP Number]:** The unique identifier for the supplementary release, corresponding to the AIP SUP (Supplementary) numbering system, this follows the format **SUP[SUP Number]** (e.g., SUP2501 for the first supplementary release of 2025).
  - **[Version Number]**: Indicates the release version of the supplementary package, formatted as **V[Version Number]**. The initial release is uniformly "V0".

- **Example:**

  CN_AIP-DS_SUP2501_V0.zip

3. Naming Conventions for Data Files

Data files within Baseline and Supplementary Packages follow a structured naming convention to clearly indicate their content, type, and version. This ensures traceability and accurate interpretation of individual files.

- **Naming Convention:**

  CN_AIP-DS_[Data Subset]_[Released Data Type]_[Effective Date]_[AIRAC or SUP Number]_[Version]_[CORRIGENDA Version].xml

- **Component Specifications:**
  - **[Data Subset]:** Specifies the subdivided content of the dataset, organized by AIXM Features. Valid values include: **Airspace, Route, RouteSegment, DesignatedPoint, AirportHeliport, Runway, RunwayDirection, TouchDownLiftOff, Navids, AeronauticalGroundLight, HoldingPattern**.
  - **[Released Data Type]:** Indicates the category of data contained in the file. Valid values:
    - **BASELINE:** when the file contains a complete AIP data set or a subset thereof.
    - **DIFF-BL**: when the the file contains an AIP Data Set Differences variant coded with BASELINE TimeSlices.
    - **DIFF-DELTA**: when the file contains an AIP Data Set Differences variant coded with PERMDELTA TimeSlices.
    - **SUP**: when the file contains a Supplementary AIP Data set.

- **[Effective Date]:** The effective time of the data file, formatted as **EFF[Time or Description]**. For time-specific validity, UTC is used (e.g., EFF202512031600). For supplementary data without a fixed effective time, business-specific labels are used (e.g., EFFbyATC, EFFWIE, EFFbyNOTAM).
- **[AIRAC or SUP Number]**: Indicates the release context:
  - **AIRAC**: For regular AIRAC-compliant releases.
  - **AIRAC-PART**: For partial releases within an AIRAC cycle.
  - **[SUP Number]**: For non-AIRAC or supplementary release numbers (e.g., NR2503).
- **[Version]**: Indicates the version of the data file, formatted as **V[Version Number]**. The initial release is "V0".
- **[CORRIGENDA Version]:** Indicates if the file is a correction patch (aligned with CORRIGENDA definitions), formatted as **CORRIGENDA-R[Version Number]**. The initial correction is "CORRIGENDA-R0".

- **Examples:**

  CN_AIP-DS_Airspace_BASELINE_EFF202512031600_AIRAC_V0.xml

  CN_AIP-DS_Airspace_DIFF-BL_EFF202512031600_AIRAC_V0.xml

  CN_AIP-DS_Airspace_BASELINE_EFF202512031600_AIRAC_V0_CORRIGENDA-R0.xml

  CN_AIP-DS_Airspace_SUP_EFFbyATC_NR2503_V0.xml

Annex D: Checklist and Data Validation

1. Overview

To ensure the integrity and validity of aeronautical data throughout its release and distribution lifecycle, a two-tiered validation mechanism is implemented. This scheme includes **package-level validation** (via a checklist.xml file) and **file-level validation** (via a checksum.xml file for each package). Both validation files are formatted in XML and adhere to standardized schemas (XSD), enabling automated verification and consistency across systems.

2. Validation Mechanisms

2.1 Package-Level Validation (checklist.xml)

This validation layer focuses on maintaining a trusted record of **all current and valid** data packages (both Baseline and Supplementary). The **checklist.xml** file serves as an authoritative manifest, ensuring that only active, unaltered packages are used in operational workflows.

- **Purpose:** To verify the status of valid data packages (e.g., identifying the latest releases, confirming active versions, and ensuring packages have not been tampered with).
- **Content:** Lists all currently valid data packages, each accompanied by:
  - Core metadata (unique ID, product type, release time, effective time, version, etc.) to uniquely identify the package.
  - The package filename (ZIP archive).
  - A SHA-256 checksum of the entire package, enabling verification of the package's integrity (e.g., detecting corruption or unauthorized modifications).
- **Technical Specification:**
  - XSD File: dataset/checklist.xsd.
  - Filename: checklist.xml.

2.2 File-Level Validation (checksum.xml)

Each data package (Baseline or Supplementary) includes a **checksum.xml** file to validate the integrity of individual files within the package. This supports end-users in verifying that all contents of the package are complete and

unaltered after download or transmission.

- **Purpose:** To confirm that every file inside a data package (e.g., XML data files) is intact and matches the state when the package was released.
- **Content:** For the specific data package, lists all contained files with:
  - Core metadata of the package (consistent with checklist.xml for traceability).
  - Filenames of individual files within the package.
  - SHA-256 checksums for each file, enabling verification of individual file integrity.
- **Technical Specification:**
  - XSD File: dataset/checksum.xsd.
  - Filename: checksum.xml (included within each data package).

3. Business Workflow

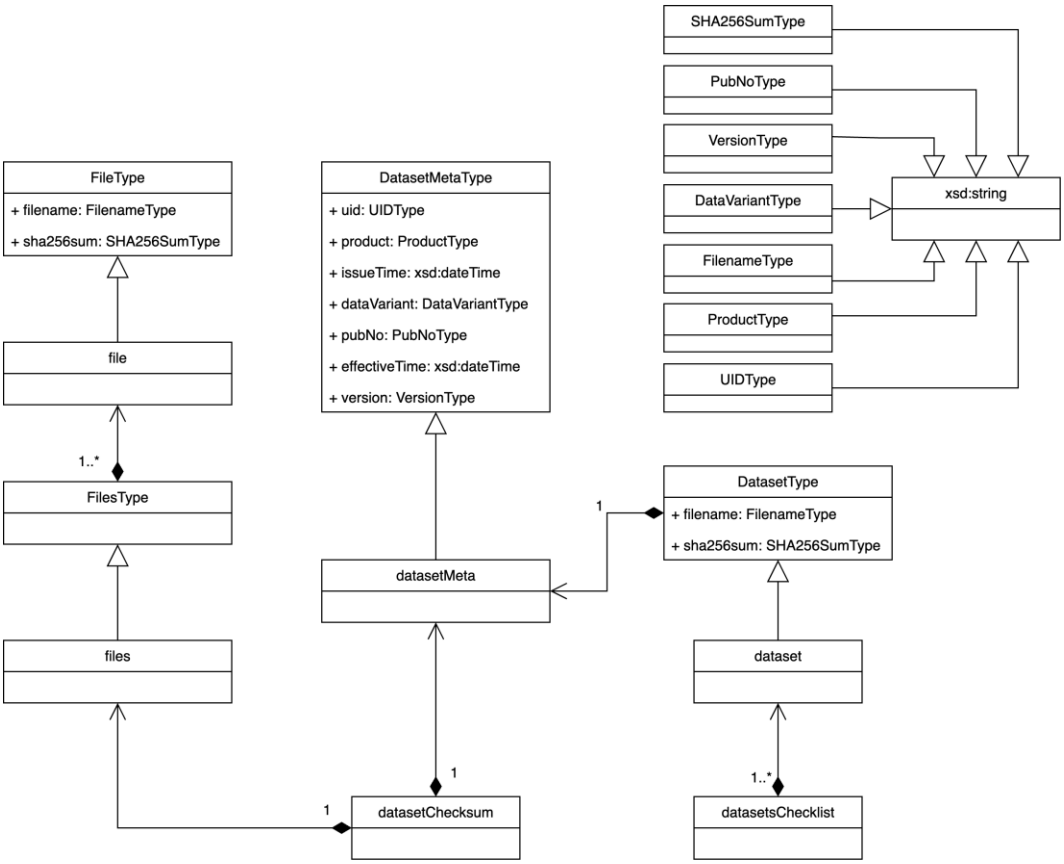
- **Release Process:**

  When a data package (Baseline or Supplementary) is published:

  1) A checksum.xml file is generated and included inside the package, containing checksums for all files within it.
  2) The package is added to the checklist.xml, which is maintained as a centralized record of all valid packages, with its SHA-256 checksum appended.
- **Verification Workflow:**
  - System/Platform Validation: Use checklist.xml to confirm that the latest valid packages are being used and that they have not been tampered with.
  - End-User Validation: After downloading a package, use the included checksum.xml to verify that all internal files are complete and unaltered.

4. Schema References

The structural models of the validation files (including metadata fields, data types, and relationships) are illustrated in the UML diagram below.



For complete technical details, refer to the corresponding XSD files:

- dataset/checklist.xsd: https://www.eaipchina.cn/schema/dataset/checksum.xsd.
- dataset/checksum.xsd: https://www.eaipchina.cn/schema/dataset/checksum.xsd.
- dataset/meta.xsd: https://www.eaipchina.cn/schema/dataset/meta.xsd.